

Practical 3 on WAR File

```
C:\Users\charl\docker-demo>mkdir web-demo
```

```
C:\Users\charl\docker-demo>cd web-demo
```

```
C:\Users\charl\docker-demo\web-demo>notepad index.jsp
```

```
<html>
<body>
<h1>Hello! This webpage is running using Maven.</h1>
</body>
</html>
```

```
C:\Users\charl\docker-demo\web-demo>notepad pom.xml
```

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.demo</groupId>
  <artifactId>web-demo</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>war</packaging>

  <name>web-demo Maven Webapp</name>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <finalName>web-demo</finalName>

    <plugins>

      <!-- FIX: Skip web.xml requirement -->
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-war-plugin</artifactId>
        <version>3.5.1</version>
        <configuration>
          <failOnMissingWebXml>false</failOnMissingWebXml>
        </configuration>
      </plugin>

      <!-- Tomcat plugin to run app -->
      <plugin>
        <groupId>org.apache.tomcat.maven</groupId>
        <artifactId>tomcat7-maven-plugin</artifactId>
        <version>2.2</version>
        <configuration>
          <port>8080</port>
          <path>/</path>
        </configuration>
      </plugin>

    </plugins>
  </build>
</project>
```

```

C:\Users\charl\docker-demo\web-demo>dir
Volume in drive C is OS
Volume Serial Number is 5088-97B8

Directory of C:\Users\charl\docker-demo\web-demo

03/31/2026  12:20 PM    <DIR>          .
03/31/2026  12:19 PM    <DIR>          ..
03/31/2026  12:20 PM                122 index.jsp
03/31/2026  12:21 PM            1,116 pom.xml
                2 File(s)            1,238 bytes
                2 Dir(s)  93,531,586,560 bytes free

```

```

C:\Users\charl\docker-demo\web-demo>mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:web-demo >-----
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ web-demo ---
[INFO] Deleting C:\Users\charl\docker-demo\web-demo\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\charl\docker-demo\web-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ web-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\charl\docker-demo\web-demo\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ web-demo ---
[INFO] No tests to run.
[INFO]
[INFO] --- war:3.5.1:war (default-war) @ web-demo ---
[INFO] Packaging webapp
[INFO] Assembling webapp [web-demo] in [C:\Users\charl\docker-demo\web-demo\target\web-demo]
[INFO] Processing war project
[INFO] Building war: C:\Users\charl\docker-demo\web-demo\target\web-demo.war
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.pom (2.2 kB at 1.8 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver/1.9.22/maven-resolver-1.9.22.pom (23 kB at 91 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.pom (2.2 kB at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.pom (7.8 kB at 113 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/17/plexus-utils-17.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/17/plexus-utils-17.pom (28 kB at 274 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.22/maven-resolver-util-1.9.22.jar (196 kB at 275 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.22/maven-resolver-api-1.9.22.jar (157 kB at 234 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 160 kB/s)
[INFO] Installing C:\Users\charl\docker-demo\web-demo\pom.xml to C:\Users\charl\.m2\repository\com\demo\web-demo\1.0-SNAPSHOT\web-demo-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\charl\docker-demo\web-demo\target\web-demo.war to C:\Users\charl\.m2\repository\com\demo\web-demo\1.0-SNAPSHOT\web-demo-1.0-SNAPSHOT.war
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 11.209 s
[INFO] Finished at: 2026-03-31T12:35:37+05:30
[INFO] -----

```

```
C:\Users\charl\docker-demo\web-demo>mvn tomcat7:run
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/maven-metadata.xml (427 B at 297 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin-2.2.pom (12 kB at 47 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin-2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat-maven-plugin-2.2.pom (33 kB at 32 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/12/apache-12.pom (16 kB at 49 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin-2.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/maven/tomcat7-maven-plugin-2.2.jar (114 kB at 135 kB/s)
[INFO]
[INFO] -----< com.demo.web-demo >-----
[INFO] Building web-demo Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] >>> tomcat7:2.2:run (default-cli) > process-classes @ web-demo >>>
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ web-demo ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\charl\docker-demo\web-demo\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ web-demo ---
[INFO] No sources to compile
[INFO]
[INFO] <<< tomcat7:2.2:run (default-cli) < process-classes @ web-demo <<<
[INFO]
[INFO]
[INFO] --- tomcat7:2.2:run (default-cli) @ web-demo ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/embed/tomcat-embed-core/7.0.47/tomcat-embed-core-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/embed/tomcat-embed-core/7.0.47/tomcat-embed-core-7.0.47.pom (1.9 kB at 27 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-util/7.0.47/tomcat-util-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-util/7.0.47/tomcat-util-7.0.47.pom (1.8 kB at 29 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-juli/7.0.47/tomcat-juli-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-juli/7.0.47/tomcat-juli-7.0.47.pom (1.3 kB at 13 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-api/7.0.47/tomcat-api-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-api/7.0.47/tomcat-api-7.0.47.pom (1.6 kB at 22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-servlet-api/7.0.47/tomcat-servlet-api-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-servlet-api/7.0.47/tomcat-servlet-api-7.0.47.pom (1.9 kB at 21 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-coyote/7.0.47/tomcat-coyote-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-coyote/7.0.47/tomcat-coyote-7.0.47.pom (1.9 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jdbc/7.0.47/tomcat-jdbc-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jdbc/7.0.47/tomcat-jdbc-7.0.47.pom (1.5 kB at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-dbcp/7.0.47/tomcat-dbcp-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-dbcp/7.0.47/tomcat-dbcp-7.0.47.pom (1.3 kB at 3.4 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jsp-api/7.0.47/tomcat-jsp-api-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jsp-api/7.0.47/tomcat-jsp-api-7.0.47.pom (2.1 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-el-api/7.0.47/tomcat-el-api-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-el-api/7.0.47/tomcat-el-api-7.0.47.pom (1.3 kB at 21 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jasper/7.0.47/tomcat-jasper-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jasper/7.0.47/tomcat-jasper-7.0.47.pom (2.8 kB at 9.9 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jdt/core/compiler/ecj/4.2.2/ecj-4.2.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jdt/core/compiler/ecj/4.2.2/ecj-4.2.2.pom (1.5 kB at 6.4 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jasper-el/7.0.47/tomcat-jasper-el-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-jasper-el/7.0.47/tomcat-jasper-el-7.0.47.pom (1.6 kB at 23 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-catalina/7.0.47/tomcat-catalina-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-catalina/7.0.47/tomcat-catalina-7.0.47.pom (2.4 kB at 14 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-annotations-api/7.0.47/tomcat-annotations-api-7.0.47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/tomcat/tomcat-annotations-api/7.0.47/tomcat-annotations-api-7.0.47.pom (1.3 kB at 13 kB/s)
```